

EDUCATION

B.Tech. in Mechanical Engineering, Department **Rank 3**
National Institute of Technology (NIT), Calicut

Dec 2020 – May 2024

- Cumulative GPA: **9.25/10.0**
- Relevant Coursework: Machine Learning, Statistics, Probability, Computer Programming

FULL-TIME WORK EXPERIENCE

Decision Analytics Associate

ZS Associates, Pune

Data Science and Advanced Analytics for Pharmaceutical Applications

Feb 2025 – Present

- Conducted marketing mix modeling for J&J's immunology brands enabling **data-driven omnichannel strategy optimization**
- Built **ETL pipelines** from ingestion of raw promotional data from various tables using **AWS redshift**, to various cleaning and aggregations using **Pandas** following the medallion architecture
- Developed **Linear** and **Bayesian regression** models with engineered features (e.g., ad stocking, normalization, log transforms) to accurately account for the impact of each marketing channel on total impactable sales
- Translated model outputs into deck-ready **insights** and excelled in **client communication**

Condition Monitoring Engineer

Hindustan Petroleum (HP), Visakhapatnam

Engineering for Large-Scale Critical Equipment Control and Monitoring

Aug 2024 – Jan 2025

- Conducted equipment monitoring using software like Honeywell Forge and Bentley Nevada System

TECHNICAL SKILLS

- **Tech Stack:** Python, SQL, Databricks, Algorithms, Excel
- **Frameworks:** PyTorch, XGBoost, PySpark, Pandas, NumPy

SELECTED PROJECTS

LLM Style Transfer via Parameter-Efficient Fine-Tuning

Personal Project, Feb 2026

Link: [GitHub](#)

- Fine-tuned **LiquidAI LFM2-1.2B** using **LoRA-based parameter-efficient fine-tuning** on 2,048 style-transfer examples to control generation style while training <0.05% of model parameters.
- Evaluated outputs using LLM-as-a-Judge methodology using **Gemini-2.5-Flash**, and **negative Log likelihood** metrics.
- Built **PyTorch** + HuggingFace pipelines for training, fine-tuning, and evaluation.

LSTM Based Real-time Music Generation Application

Personal Project, Jan 2026

Link: [Live Demo](#), [GitHub](#)

- Trained a character-level **LSTM** on ~800 songs with **embedding**, **sequence batching**, and **multinomial sampling** with **temperature control** to learn **next-token probability** distributions for symbolic music generation.
- **Deployed** a **Streamlit**-based interactive **inference** system using pretrained weights with configurable generation parameters, real-time audio synthesis, and live web deployment.

Promotion Impact Estimation using Bayesian Regression

Decision Analyst, ZS, Dec 2025

- Built **Bayesian regression** models to quantify channel-level contribution across marketing channels for a rapidly growing J&J's immunology brand.
- Delivered **actionable insights** enabling **data-driven** budget allocation, maximizing **ROI** and optimizing spend efficiency in competitive immunology markets.

OTHER RELEVANT COURSES

- **Verified Certifications:** Databricks Certified - Data Engineer Associate, [Link](#)
- **Online Courses:** Stanford Online - Deep Learning Certificate, [Link](#)

PUBLICATIONS

- **Vasanth B S**, Vysakh P, M Thashreef, A P Sudheer. "Design, Fabrication and Control of a Multi-purpose Marine Robot", NIT Calicut, [Paper](#) (To be Submitted)

May 2024